ONE PREFERENCE CHANGE ON EACH SIDE: WHICH ROBUST SUBLATTICES OCCUR

JI HO BAE

Abstract. Let two stable-marriage profiles A, B differ in at most one worker's list and at most one firm's list. Put $L = \mathcal{M}_A$ and $R = \mathcal{M}_A \cap \mathcal{M}_B$. We determine exactly which pairs (L, R) occur, up to lattice isomorphism. Writing L as the down-set lattice of its join-irreducibles P, the occurring sets R are exactly those cut out by two chains of P: an arbitrary set of admissible positions on each chain, and implications whose tail lies on the first chain or whose head lies on the second. In particular, not every sublattice occurs. The complement $L \setminus R$ can be closed under meets, under joins, under both, or under neither.

1. Main result

Throughout, W and F are equal-sized finite sets of workers and firms, and profiles are strict and complete. A matching M is stable for a profile if no worker–firm pair strictly prefers each other to their partners. The stable matchings $\mathcal{M}_A$ form a distributive lattice under unanimous weak worker preference, the matching better for workers being smaller.

For a finite poset P let $\mathcal{D}(P)$ be the lattice of its down-sets under inclusion. Birkhoff's theorem gives $L \cong \mathcal{D}(J(L))$, where $J(L)$ is the poset of join-irreducible elements.

Definition 1. A *one-change system* on P consists of chains $C_1 = (\alpha_1 < \cdots < \alpha_p)$ and $C_2 = (\beta_1 < \cdots < \beta_q)$ of P (either may be empty), sets $A_1 \subseteq \{0, \dots, p\}$ and $A_2 \subseteq \{0, \dots, q\}$, and a finite set Φ of implications $s \to t$ of two kinds:

$$(w\text{-kind}) \quad s \in C_1 \cup \{\top\},\ t \in P \cup \{\bot\}; \qquad (f\text{-kind}) \quad s \in P \cup \{\top\},\ t \in C_2 \cup \{\bot\}.$$

An implication can be of both kinds. A down-set X *satisfies* $s \to t$ if $s \in X$ implies $t \in X$, where $\top \in X$ and $\bot \notin X$ by convention. The solution set is

$$\mathrm{Sol} = \{X \in \mathcal{D}(P) :\ |X \cap C_1| \in A_1,\ |X \cap C_2| \in A_2,\ X \text{ satisfies every member of } \Phi\}.$$

The maps $X \mapsto |X \cap C_i|$ are lattice homomorphisms onto chains, and each implication defines a sublattice, so every solution set is a sublattice of $\mathcal{D}(P)$ or empty.

Theorem 2. *Let L be a finite distributive lattice, $P = J(L)$, and $R \subseteq L$. There are profiles A, B, differing in at most one worker's list and at most one firm's list, and a lattice isomorphism $\mathcal{M}_A \to L$ carrying $\mathcal{M}_A \cap \mathcal{M}_B$ onto R, if and only if R corresponds, under $L \cong \mathcal{D}(P)$, to the solution set of a one-change system on P.*

Example 3. Let $P = \{x_1, x_2, x_3, x_4\}$ be an antichain, so $L \cong \{0,1\}^4$, and let $R = \{X :\ x_1 \in X \Leftrightarrow x_2 \in X,\ x_3 \in X \Leftrightarrow x_4 \in X\}$. This is a sublattice, but it does not occur. Suppose a system has $\mathrm{Sol} = R$. Every chain of P has at most one element, and R contains $\varnothing$ and P, so A_1 and A_2 contain every position and exclude nothing. The only nontrivial implications valid on R are $x_1 \to x_2$, $x_2 \to x_1$, $x_3 \to x_4$ and $x_4 \to x_3$, and each is needed: the down-set $\{x_1\}$ violates only the first, and so on. An admissible implication has its tail in C_1 or its head in C_2. One element of C_1 is the tail of only one of the four, and one element of C_2 is the head of only one. So at most two are admissible, and $\mathrm{Sol} \neq R$.

Date: October 10, 2026.

Corollary 4. *Put $D = L \setminus R$. Among the pairs that occur, D can be meet-closed or not and join-closed or not, in all four combinations. So these properties are not forced by realizability and must be checked on (L, R) directly.*

Proof. Take $P = \{x, y\}$ an antichain and $C_1 = C_2 = \{x\}$, with all positions allowed. The implications $x \to y$ (tail in C_1) and $y \to x$ (head in C_2) give $R = \{\varnothing, \{x, y\}\}$, whose complement is closed under neither operation. The implications $\top \to x$ and $\top \to y$ give $R = \{\{x, y\}\}$, whose complement is meet-closed but not join-closed. The implications $x \to \bot$ and $y \to \bot$ give $R = \{\varnothing\}$, whose complement is join-closed but not meet-closed. Finally $P = \{x\}$ with $C_1 = C_2 = \{x\}$ and $\top \to x$ gives a complement $\{\varnothing\}$ closed under both. □

2. Rotations

We use the rotation theory of Gusfield and Irving [2, Ch. 2–3]. Fix a profile A, and let M_0 be its worker-optimal stable matching, which worker-proposing deferred acceptance returns [1]. For a stable M and a worker x, the successor $s_M(x)$ is the first firm after $M(x)$ on x's list that prefers x to its partner, if one exists; put $\mathrm{next}_M(x) = M^{-1}(s_M(x))$. A *rotation exposed in M* is the cyclic sequence of pairs $(x_i, M(x_i))$ along a cycle $(x_0, \dots, x_{r-1})$ of next_M, and eliminating it gives each x_i the partner $M(x_{i+1})$. We use these facts.

(R1) Eliminating an exposed rotation from a stable matching gives a stable matching, and every stable matching is reached from M_0 by successive eliminations.

(R2) Let Π be the set of rotations, ordered by precedence. Then $X \mapsto M_X$, where M_X is M_0 with the rotations of X eliminated in any order compatible with precedence, is a lattice isomorphism $\mathcal{D}(\Pi) \to \mathcal{M}_A$. The rotations exposed in M_X are the minimal elements of $\Pi \setminus X$. Hence $\Pi \cong J(\mathcal{M}_A)$.

(R3) For an agent a, let $C_a \subseteq \Pi$ consist of the rotations whose pairs contain a. Then C_a is a chain, and $M_X(a)$ depends only on $|X \cap C_a|$. Along C_a a worker's partner strictly worsens and a firm's strictly improves.

For (R3): eliminating a rotation changes exactly the partners of its members, a worker moving down its list and a firm up. If $\rho, \sigma \in C_a$ were incomparable, let X be the union of their strict down-sets. Both would be minimal in $\Pi \setminus X$, hence exposed in M_X by (R2), and both would contain a. But next_{M_X} is a function, so distinct exposed rotations share no worker, and hence no firm, since the firms of a rotation are the partners of its workers. Rotations outside C_a do not move a, so $M_X(a)$ depends only on $X \cap C_a$, an initial segment of C_a determined by its size.

(R4) [3, Theorem 7] If A, B differ in at most one worker's list (any number of firms may change), then $\mathcal{M}_A \cap \mathcal{M}_B$ is closed under the meet and join of $\mathcal{M}_A$.

For completeness, here is a short proof of (R4). Let M, M' be stable in both profiles. On each cycle of $M \cup M'$ all workers prefer the same one of M, M' in a given profile: if x prefers $y = M(x)$ to $M'(x)$, stability of M' makes y prefer $x_2 = M'(y)$ to x, and stability of M makes x_2 prefer $M(x_2)$ to y, and so on around the cycle. A cycle through w also contains another worker, whose list is the same in A and B. So the meets of M, M' in A and in B coincide, and the meet lies in $\mathcal{M}_A \cap \mathcal{M}_B$. Joins are similar.

3. Necessity

Let A, B differ at most in the lists of a worker w and a firm f. If fewer lists differ, choose w or f arbitrarily. Use (R2) to identify $\mathcal{M}_A$ with $\mathcal{D}(\Pi)$, and let

$$R^* = \{X \in \mathcal{D}(\Pi) : \ M_X \in \mathcal{M}_B\}.$$

Put $C_1 = C_w = (\alpha_1 < \cdots < \alpha_p)$ and $C_2 = C_f = (\beta_1 < \cdots < \beta_q)$, and write $\pi_1(X) = |X \cap C_1|$ and $\pi_2(X) = |X \cap C_2|$. Let p_i be w's partner when $\pi_1 = i$, and q_j be f's partner when $\pi_2 = j$. By (R3), $p_0 \succ_w^A p_1 \succ_w^A \cdots$ and $q_0 \prec_f^A q_1 \prec_f^A \cdots$. Write $\alpha_0 = \top$ and $\beta_{q+1} = \bot$.

If $R^* = \varnothing$, take $A_1 = \varnothing$, any A_2 and $\Phi = \varnothing$. Otherwise let $A_1 = \pi_1(R^*)$ and $A_2 = \pi_2(R^*)$, and let Φ be the set of all implications of either kind that every member of R^* satisfies. Then $R^* \subseteq \mathrm{Sol}$. We prove $\mathrm{Sol} \subseteq R^*$.

Thresholds. For a firm $g \neq f$, the partners of g improve along C_g, so the down-sets X with $w \succ_g^A M_X(g)$ are those with $|X \cap C_g| < \tau$ for some τ. Accordingly there is $y_g \in C_g \cup \{\top, \bot\}$ with

$$w \succ_g^A M_X(g) \iff y_g \notin X \qquad (X \in \mathcal{D}(\Pi)).$$

Symmetrically, for a worker $v \neq w$ there is $z_v \in C_v \cup \{\top, \bot\}$ with $f \succ_v^A M_X(v) \iff z_v \in X$. For $g = p_k \neq f$ with $k \geq 1$ we have $y_{p_k} = \alpha_k$, because p_k receives w exactly when α_k is eliminated. Likewise $z_{q_l} = \beta_{l+1}$ for $l < q$ with $q_l \neq w$, because q_l holds f or a better firm until β_{l+1} is eliminated, which moves it from f to a worse firm.

Lemma 5. *Let $i \in A_1$ and $i < k \leq p$ with $p_k \neq f$. Then $p_i \succ_w^B p_k$. Let $j \in A_2$ and $0 \leq l < j$ with $q_l \neq w$. Then $q_j \succ_f^B q_l$.*

Proof. Take $Y \in R^*$ with $\pi_1(Y) = i$. Then $\alpha_k \notin Y$, so $w \succ_{p_k} M_Y(p_k)$, and p_k's list is unchanged. If $p_k \succ_w^B p_i$, the pair (w, p_k) would block M_Y in B. The second claim is symmetric, using Y with $\pi_2(Y) = j$, so that $\beta_{l+1} \in Y$. □

Lemma 6. *Suppose $M_{X_0}(w) = f$ for some X_0. Then there are j_0, j_1 such that, for every $X \in \mathcal{D}(\Pi)$, $M_X(w) = f$ iff $\pi_1(X) = j_0$ iff $\pi_2(X) = j_1$, and $\pi_1(X) \geq j_0$ iff $\pi_2(X) \geq j_1$.*

Proof. Let $f = p_{j_0}$ and $w = q_{j_1}$. If $j_0 = 0$ then w, f are partners in M_0, so $j_1 = 0$ and the last claim is trivial. If $j_0 \geq 1$, the rotation α_{j_0} moves w onto f, so it changes f's partner to w and equals β_{j_1}. Thus $\pi_1(X) \geq j_0 \iff \alpha_{j_0} \in X \iff \pi_2(X) \geq j_1$. □

Proof of $\mathrm{Sol} \subseteq R^*$. Let $X \in \mathrm{Sol}$, $i = \pi_1(X)$ and $j = \pi_2(X)$, and suppose a pair blocks M_X in B. Pairs avoiding w and f have their A-preferences, and M_X is A-stable, so the pair contains w or f. In each case below we exhibit an implication of Φ's kind that X violates, and derive a contradiction from its failure on R^*; hence it lies in Φ, contradicting $X \in \mathrm{Sol}$.

(a) The pair is (w, g) with $g \neq f$. Then $g \succ_w^B p_i$ and $y_g \notin X$, so $y_g \neq \top$. The implication $\alpha_i \to y_g$ is of w-kind and X violates it. Suppose $Y \in R^*$ violates it too, and take $Y' \in R^*$ with $\pi_1(Y') = i$. By (R4), $Z = Y \cap Y' \in R^*$. Since $\alpha_i \in Y$ we get $\pi_1(Z) = i$, and $y_g \notin Z$. So (w, g) blocks M_Z in B, a contradiction.

(b) The pair is (v, f) with $v \neq w$. Then $v \succ_f^B q_j$ and $z_v \in X$, so $z_v \neq \bot$. The implication $z_v \to \beta_{j+1}$ is of f-kind and X violates it. If $Y \in R^*$ violates it, take $Y' \in R^*$ with $\pi_2(Y') = j$ and $Z = Y \cup Y'$. Then $\pi_2(Z) = j$ and $z_v \in Z$, so (v, f) blocks M_Z in B.

(c) The pair is (w, f). Then $M_X(w) \neq f$, $f \succ_w^B p_i$ and $w \succ_f^B q_j$. The implication $\alpha_i \to \beta_{j+1}$ is of both kinds and X violates it. Let $Y \in R^*$ violate it, with $k = \pi_1(Y) \geq i$ and $l = \pi_2(Y) \leq j$. If $M_Y(w) \neq f$, then $p_k \neq f$ and $q_l \neq w$, and Lemma 5 gives $p_i \succeq_w^B p_k$ and $q_j \succeq_f^B q_l$ (the first trivially if $k = i$, the second if $l = j$). Hence $f \succ_w^B p_k$ and $w \succ_f^B q_l$, and (w, f) blocks M_Y. If $M_Y(w) = f$, Lemma 6 gives $k = j_0$ and $l = j_1$. Then $\pi_2(X) = j \geq j_1$ forces $i \geq j_0 = k \geq i$, so $\pi_1(X) = j_0$ and $M_X(w) = f$, a contradiction. □

So $R^* = \mathrm{Sol}$. A lattice isomorphism $\mathcal{M}_A \to L$ carrying $\mathcal{M}_A \cap \mathcal{M}_B$ onto R restricts to a poset isomorphism $\Pi \cong J(\mathcal{M}_A) \to J(L) = P$. It carries our system on Π to a one-change system on P whose solution set corresponds to R. This proves necessity.

4. Sufficiency

Fix a finite poset P and a one-change system on it with chains $C_1 = (\alpha_1 < \cdots < \alpha_p)$ and $C_2 = (\beta_1 < \cdots < \beta_q)$. We build A and B.

Agents. The workers are m_ρ, m'_ρ for $\rho \in P$, together with $w, c_1, \dots, c_p, d_0, d_1, \dots, d_q, u, v$. The firms are h_ρ, h'_ρ for $\rho \in P$, together with $e_0, \dots, e_p, f, k_1, \dots, k_q, g, \hat{f}$.

Cycles. For $\rho \in P$ let Γ_ρ be the cyclic sequence of workers formed by concatenating (w, c_t) if $\rho = \alpha_t$, then (d_s, d_{s-1}) if $\rho = \beta_s$, then (m_ρ, m'_ρ). The initial matching is

$$M_\varnothing = \{m_\rho h_\rho,\ m'_\rho h'_\rho\ (\rho \in P),\ we_0,\ c_t e_t,\ d_0 f,\ d_s k_s\ (s \geq 1),\ ug,\ v\hat{f}\}.$$

For $X \in \mathcal{D}(P)$, the matching M_X is obtained from $M_\varnothing$ by *eliminating* the members of X in an order compatible with P, where eliminating ρ gives each worker of Γ_ρ the current partner of the next worker of Γ_ρ. Two cycles share a worker only if both rotations lie in C_1 or both lie in C_2, hence only if they are comparable. Eliminations along disjoint cycles commute, and any two orders compatible with P differ by swaps of adjacent incomparable elements, so M_X does not depend on the order chosen.

Each m_ρ, m'_ρ moves only at ρ, each c_t only at α_t, each d_s only at β_s and β_{s+1}, and w at $\alpha_1, \dots, \alpha_p$ in turn. So each agent's partners are as follows. Put $\kappa_t = k_s$ if $\alpha_t = \beta_s$, and $\kappa_t = h_{\alpha_t}$ otherwise. Put $\varepsilon_\rho = e_{t-1}$ if $\rho = \alpha_t$; otherwise $\varepsilon_\rho = k_s$ if $\rho = \beta_s$; otherwise $\varepsilon_\rho = h_\rho$. Put $\lambda_\rho = d_{s-1}$ if $\rho = \beta_s$; otherwise $\lambda_\rho = c_t$ if $\rho = \alpha_t$; otherwise $\lambda_\rho = m'_\rho$.

- Workers, in order of occurrence: w: $e_0, e_1, \dots, e_p$. c_t: e_t, κ_t. d_s: k_s (if $s \geq 1$), f, $h_{\beta_{s+1}}$ (if $s < q$). m_ρ: h_ρ, h'_ρ. m'_ρ: $h'_\rho, \varepsilon_\rho$. u: g. v: $\hat{f}$.
- Firms, in order of occurrence: e_t: c_t (if $t \geq 1$), w, $m'_{\alpha_{t+1}}$ (if $t < p$). f: $d_0, \dots, d_q$. k_s: d_s, then c_t if $\beta_s = \alpha_t$ and m'_{β_s} otherwise. h_ρ: m_ρ, λ_ρ. h'_ρ: m'_ρ, m_ρ. g: u. $\hat{f}$: v.

Profile A. A worker lists its partners in order of occurrence and a firm in reverse order, so each agent prefers the later partners if it is a firm and the earlier ones if it is a worker. Then make these insertions.

(I1) For each covering pair $\sigma \lessdot \tau$ of P: h_σ into m_τ's list directly after h_τ, and m_τ into h_σ's list directly before m_σ.
(I2) For each $y \in P$ that is the head of a w-kind implication: w into h_y's list directly before m_y.
(I3) For each $z \in P$ that is the tail of an f-kind implication: f into m_z's list directly after h_z.
(I4) g's list begins w, u, and v's list begins $f, \hat{f}$.

Insertions at the same place are made in any order. All remaining agents follow, in any order, at the end of every list. Call these the *tails* of the lists.

Lemma 7. *The stable matchings of A are exactly the M_X with $X \in \mathcal{D}(P)$. They are distinct, and $X \mapsto M_X$ is a lattice isomorphism $\mathcal{D}(P) \to \mathcal{M}_A$. Moreover $M_X(w) = e_{|X \cap C_1|}$ and $M_X(f) = d_{|X \cap C_2|}$; h_y holds m_y iff $y \notin X$; and m_z holds h_z iff $z \notin X$.*

Proof. In worker-proposing deferred acceptance every worker first proposes to its first choice. These are distinct except that d_0 and v both propose to f, which keeps d_0 because v lies in its tail. Then v is accepted by $\hat{f}$. So $M_\varnothing$ is the worker-optimal stable matching.

Fix $X \in \mathcal{D}(P)$. A worker is *finished* if it holds its last partner. For an unfinished worker x, let $\mathrm{cur}(x)$ be the least member of $C_x \setminus X$, where now $C_x = \{\rho : x \in \Gamma_\rho\}$; this is the rotation that next moves x. We compute $\mathrm{next} = \mathrm{next}_{M_X}$. A firm prefers x to its partner only if x is above that partner on its list. Before its tail, every firm lists only its partners, the insertions (I1)–(I4), and nothing else, and every firm lists its last partner first, except g. Using this, the successors are as follows.

(i) $x = m_\tau$, $\tau \notin X$. It holds h_τ. Next come, in some order, the firms h_σ for $\sigma \lessdot \tau$ and possibly f, then h'_τ. The firm f keeps m_τ in its tail. The firm h_σ prefers m_τ iff it holds m_σ, i.e. iff $\sigma \notin X$. If some such $\sigma \notin X$, then $\mathrm{next}(x) = m_\sigma$ for the first such σ on the list, and $\mathrm{cur}(m_\sigma) = \sigma < \tau$. Otherwise $\mathrm{next}(x) = m'_\tau$, and every lower cover of τ, hence everything below τ, lies in X.

(ii) $x = m'_\tau$, $\tau \notin X$. It holds h'_τ and its next entry is ε_τ, which lists m'_τ first. If $\tau = \alpha_t$, the holder of e_{t-1} is w when $\alpha_{t-1} \in X$ (or $t = 1$), with $\mathrm{cur}(w) = \tau$; otherwise it is c_{t-1}, with $\mathrm{cur}(c_{t-1}) = \alpha_{t-1} < \tau$. If $\tau = \beta_s \notin C_1$, the holder of k_s is d_s, with $\mathrm{cur}(d_s) = \tau$. Otherwise the holder of h_τ is m_τ.

(iii) $x = w$ holding e_i with $i < p$. Its next entry e_{i+1} holds c_{i+1} and prefers w, so $\mathrm{next}(w) = c_{i+1}$ and $\mathrm{cur}(c_{i+1}) = \alpha_{i+1} = \mathrm{cur}(w)$.

(iv) $x = c_t$, $\alpha_t \notin X$. Its next entry κ_t lists c_t first and holds m_{α_t} or d_s, both with $\mathrm{cur} = \alpha_t$.

(v) $x = d_s$ with $s \geq 1$ and $\beta_s \notin X$. Its next entry is f, which holds d_j with $j < s$ and prefers d_s. If $j = s - 1$ then $\mathrm{cur}(d_{s-1}) = \beta_s$; otherwise $\mathrm{cur}(d_j) = \beta_{j+1} < \beta_s$.

(vi) $x = d_s$ holding f with $s < q$. Its next entry $h_{\beta_{s+1}}$ lists d_s first and holds $m_{\beta_{s+1}}$, with $\mathrm{cur} = \beta_{s+1} = \mathrm{cur}(d_s)$.

(vii) Finished workers. A finished worker $x \neq w$ has only its tail after its partner. A firm there is not a partner of x, and the only insertions of such workers into firm lists, (I1) and u in (I4), go into firms that precede their tails. So the firm lists x in its own tail, below its partner, and x has no successor. The worker w holding e_p may have successor g, held by u, or some h_y with $y \notin X$, held by m_y. No worker has w as its next, since e_p lists w first.

In each case where cur is unchanged, the new worker follows x in the cyclic order of $\Gamma_{\mathrm{cur}(x)}$. Thus every edge of next either lowers cur strictly, or joins consecutive workers of $\Gamma_{\mathrm{cur}(x)}$, or leaves w for u or for an unfinished worker. No cycle contains a finished worker, since by (vii) only w among them has a successor and no worker has w as its next. So along a cycle cur is constant, say τ, and the cycle is Γ_τ. By (i) this forces τ to be minimal in $P \setminus X$. Conversely, if τ is minimal, then (i)–(vi) supply every edge of Γ_τ: when $\tau = \alpha_t$ we have $\alpha_{t-1} \in X$, and when $\tau = \beta_s$ the firm f holds d_{s-1}.

So the rotations exposed in M_X are the Γ_τ with τ minimal in $P \setminus X$, and eliminating Γ_τ yields $M_{X \cup \{\tau\}}$. By (R1) and induction on $|X|$, every M_X is stable and these are all the stable matchings. The set X is recovered from M_X as $\{\tau : m_\tau \text{ holds } h'_\tau\}$. Workers only lose by eliminations, so $X \subseteq Y$ implies $M_X \leq M_Y$. Conversely, if $M_X \leq M_Y$ and $\tau \in X$, then m_τ holds its last partner h'_τ in M_X, hence in M_Y, so $\tau \in Y$. The remaining claims are read off the partner lists. □

Profile B. Only w and f change. Each agent below is listed only at its first occurrence. With $\alpha_0 = \top$, the list of w is: for $i = 0, 1, \ldots, p$ in turn, first the firms h_y for each $(\alpha_i \to y) \in \Phi$ with $y \in P$, and g if $(\alpha_i \to \bot) \in \Phi$, then e_i if $i \in A_1$. Next comes g if $A_1 \neq \{0, \ldots, p\}$, then the e_i with $i \notin A_1$, then everyone else. The list of f is: first the workers m_z for each $(z \to \bot) \in \Phi$ with $z \in P$, and v if $(\top \to \bot) \in \Phi$. Then, for $l = q, q-1, \ldots, 0$ in turn: d_l if $l \in A_2$, followed, when $l \geq 1$, by the workers m_z for each $(z \to \beta_l) \in \Phi$ with $z \in P$, and v if $(\top \to \beta_l) \in \Phi$. Next comes v if $A_2 \neq \{0, \ldots, q\}$, then the d_l with $l \notin A_2$, then everyone else.

Two implications with the same head y and tails $\alpha_k, \alpha_{k'}$, $k < k'$, are together equivalent to the one with tail α_k, since $\alpha_{k'} \in X$ implies $\alpha_k \in X$. Symmetrically on the f side, the head β_l with the largest l is the binding one. So listing each agent only at its first occurrence realizes all of Φ.

Lemma 8. *For $X \in \mathcal{D}(P)$, M_X is stable in B iff $X \in$ Sol.*

Proof. Let $i = |X \cap C_1|$ and $j = |X \cap C_2|$, so $M_X(w) = e_i$ and $M_X(f) = d_j$. Pairs avoiding w, f keep their A-lists and do not block. The pair (w, f) does not block, since f lies after e_i on w's new list.

Pairs (w, y) with $y \neq f$ above e_i on w's new list. If $i \notin A_1$, then g is such a firm and always prefers w to u, so (w, g) blocks; and $X \notin$ Sol. If $i \in A_1$, the firms above e_i are of three sorts. The first is e_k with $k < i$; it holds $m'_{\alpha_{k+1}}$, its first choice, so it does not block. The second is h_y listed for $(\alpha_k \to y)$ with $k \leq i$; by Lemma 7 and (I2) it prefers w iff $y \notin X$, so it blocks iff that implication fails, since $\alpha_k \in X$. The third is g listed for $(\alpha_k \to \bot)$ with $k \leq i$; it blocks, matching the failure of that implication.

Pairs (x, f) with $x \neq w$ above d_j on f's new list. If $j \notin A_2$, then v is such a worker, always prefers f to $\hat{f}$, and blocks. If $j \in A_2$, the workers above d_j are of three sorts. The first is d_l with $l > j$; it holds k_l, its first choice, so it does not block. The second is m_z listed for $(z \to \beta_l)$ with $l > j$, or for $(z \to \bot)$; by Lemma 7 and (I3), m_z prefers f to its partner iff $z \in X$, so it blocks iff the implication fails, since $\beta_l \notin X$. The third is v listed for $(\top \to \beta_l)$ with $l > j$, or for $(\top \to \bot)$, which blocks, matching the failure of that implication.

Hence M_X is B-stable iff $i \in A_1$, $j \in A_2$ and X satisfies Φ. □

Lemmas 7 and 8 prove sufficiency: A and B differ only in the lists of w and f, and the isomorphism $X \mapsto M_X$ carries Sol onto $\mathcal{M}_A \cap \mathcal{M}_B$. With necessity, this proves Theorem 2.

Use of AI tools

The author used research AI agents that the author built and configured (GPT-6 Astra max, Claude Opus 5.5 max and GPT-6 Astra Pro) during the proof work, for proof search and construction, proof checking, translation, Python computation and similar tasks. The final manuscript was written, and the proofs were checked, by the human author. All responsibility for this work rests with the author personally.

References

[1] D. Gale and L. S. Shapley, College admissions and the stability of marriage, *Amer. Math. Monthly* 69 (1962), 9–15.

[2] D. Gusfield and R. W. Irving, *The Stable Marriage Problem: Structure and Algorithms*, MIT Press, 1989.

[3] R. R. Gangam, T. Mai, N. Raju and V. V. Vazirani, Stable matching: dealing with changes in preferences, arXiv:2304.02590v3 (2025), Theorem 7 and Section 4.

LawAll AI